M378K: March 25th, 2026.

Problem 15.2. *Suppose that the two estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased. We know that* $\mathrm{Var}[\hat{\theta}_1] = \sigma_1^2$ *and* $\mathrm{Var}[\hat{\theta}_2] = \sigma_2^2$.

Consider ~~the estimator~~ all the estimators that can be obtained as convex combinations of $\hat{\theta}_1$ and $\hat{\theta}_2$, i.e., all the estimators of the form

$$\boxed{\hat{\theta} = \alpha\hat{\theta}_1 + (1-\alpha)\hat{\theta}_2.}$$

What can you say about the bias *of estimators $\hat{\theta}$ of the form above? Assuming that $\hat{\theta}_1$ and $\hat{\theta}_2$ are* **independent**, *for which weight α is the variance minimal?*

$\longrightarrow$: $\mathbb{E}[\hat{\Theta}] = \mathbb{E}[\alpha\hat{\Theta}_1 + (1-\alpha)\hat{\Theta}_2] = \alpha\overbrace{\mathbb{E}[\hat{\Theta}_1]}^{\theta} + (1-\alpha)\overbrace{\mathbb{E}[\hat{\Theta}_2]}^{\theta} = \theta$

linearity of $\mathbb{E}$

$\Rightarrow \hat{\Theta}$ is unbiased for all α

$$\mathrm{Var}[\hat{\Theta}] \xrightarrow[\alpha]{} \min$$

$$\mathrm{Var}[\alpha\hat{\Theta}_1 + (1-\alpha)\hat{\Theta}_2] \xrightarrow[\alpha]{} \min$$

independence

$$\alpha^2\,\mathrm{Var}[\hat{\Theta}_1] + (1-\alpha)^2\cdot\mathrm{Var}[\hat{\Theta}_2] \xrightarrow[\alpha]{} \min$$

$$\boxed{\alpha^2\,\sigma_1^2 + (1-\alpha)^2\cdot\sigma_2^2} \xrightarrow[\alpha]{} \min$$

$$\cancel{2}\alpha\sigma_1^2 + \cancel{2}(1-\alpha)(-1)\cdot\sigma_2^2 = 0$$

$$\alpha\sigma_1^2 + \alpha\sigma_2^2 = \sigma_2^2$$

$$\boxed{\alpha^* = \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}}$$

$\square$

Def'n. An estimator $\hat{\theta}$ of θ is said to be **linear** if it's of the form

$$\hat{\theta} = \alpha_1 Y_1 + \alpha_2 Y_2 + \cdots + \alpha_n Y_n$$

where α_i, $i = 1..n$ are all constants.

e.g., $\bar{Y}$

counterexample: sample variance

Def'n. An estimator $\hat{\theta}$ of θ is said to be

uniformly minimum variance unbiased estimator (UMVUE)

if:
- $\hat{\theta}$ is unbiased

and
- $MSE(\hat{\theta}) \leq MSE(\hat{\theta}')$ for all other unbiased estimators $\hat{\theta}'$ of θ.

Examples. These are UMVUE:

- $\bar{Y}$ for μ where $(Y_1, \ldots, Y_n)$ is a random sample from $N(\mu, \sigma)$
- $\bar{Y}$ for p where $(Y_1, \ldots, Y_n)$ is a random sample from $B(p)$
- $$\boxed{S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2}$$

  for σ^2 where $(Y_1, \ldots, Y_n)$ is a random sample from $N(\mu, \sigma)$ w/ both parameters unknown

Problem 15.3. *Let $Y_1, Y_2, \ldots, Y_n$ be a random sample from a continuous distribution with probability density function*

$$\boxed{f_Y(y) = \frac{\alpha y^{\alpha-1}}{\theta^\alpha} \mathbf{1}_{[0,\theta]}(y)}$$

with a known parameter $\alpha > 0$ and an unknown parameter $\theta > 0$. We propose the estimator $\hat{\theta} = \max(Y_1, \ldots, Y_n)$. Is this estimator unbiased*? If not, how would you modify it to create an* unbiased *estimator? What is the* **mean-squared error** *of the* unbiased *estimator you obtained?*

$\rightarrow$:

$$\hat{\theta} = \max(Y_1, \ldots, Y_n)$$

$$\mathbb{E}[\hat{\theta}] = \mathbb{E}[Y_{(n)}] = ?$$

$$\frac{\overbrace{(F_Y(y))^{n-1}}\, f_Y(y)}{y}$$

$$g_{(n)}(y) = n \cdot \boxed{(F_Y(y))}^{n-1} \cdot f_Y(y)$$

$y \in (0, \theta)$:

$$F_Y(y) = \int_0^y f_Y(u)\,du = \int_0^y \frac{\alpha u^{\alpha-1}}{\theta^\alpha}\,du = \frac{\alpha}{\theta^\alpha}\int_0^y u^{\alpha-1}\,du = \frac{\alpha}{\theta^\alpha} \cdot \frac{y^\alpha}{\alpha} = \left(\frac{y}{\theta}\right)^\alpha$$

$$\mathbb{E}[\hat{\theta}] = \mathbb{E}[Y_{(n)}] = \int_0^\theta y \cdot g_{(n)}(y)\,dy$$

$$= \int_0^\theta y \cdot n \left(\left(\frac{y}{\theta}\right)^\alpha\right)^{n-1} \cdot \frac{\alpha y^{\alpha-1}}{\theta^\alpha}\,dy$$

$$= n \cdot \alpha \cdot \frac{1}{\theta^{n\alpha}} \int_0^\theta y \cdot y^{\alpha(n-1)} \cdot y^{\alpha-1}\,dy$$

$$= \frac{n\alpha}{\theta^{n\alpha}} \int_0^\theta y^{n\alpha}\,dy = \frac{n\alpha}{\cancel{\theta^{n\alpha}}} \cdot \frac{\theta^{\cancel{n\alpha}+1}}{n\alpha+1} = \boxed{\left(\frac{n\alpha}{n\alpha+1}\right)}\,\theta$$

Not unbiased!

Let $\hat{\theta}'$ denote the unbiased modification of $\hat{\theta}$.

$$\mathbb{E}[\hat{\theta}'] = \theta$$

Set $\boxed{\hat{\theta}'} = \frac{n\alpha+1}{n\alpha} \cdot \hat{\theta} = \boxed{\frac{n\alpha+1}{n\alpha} Y_{(n)}}$

Then, $\mathbb{E}[\hat{\theta}'] = \mathbb{E}\left[\frac{n\alpha+1}{n\alpha} Y_{(n)}\right] = \frac{\cancel{n\alpha+1}}{\cancel{n\alpha}} \cdot \frac{\cancel{n\alpha}}{\cancel{n\alpha+1}} \cdot \theta = \theta$ ✓

Synopsis: $MSE[\hat{\theta}'] = ?$

1st $MSE[\hat{\theta}'] = Var[\hat{\theta}'] + \overbrace{(bias(\hat{\theta}'))^2}^{0=} = Var[\hat{\theta}']$

2nd $Var[\hat{\theta}'] = Var\left[\frac{n\alpha+1}{n\alpha} \cdot Y_{(n)}\right] = \left(\frac{n\alpha+1}{n\alpha}\right)^2 Var[Y_{(n)}]$

3rd $Var[Y_{(n)}] = \boxed{\mathbb{E}[Y_{(n)}^2]} - (\mathbb{E}[Y_{(n)}])^2$ ✓